

KUNTAL PAL

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SUMMARY

AI Engineer and PhD physicist with hands-on experience across model training, fine-tuning, retrieval systems, and production ML deployment. Passionate about pushing the frontier of efficient AI - novel architectures, inference optimization, and distillation techniques that make models faster, smaller, and more reliable in production.

TECHNICAL SKILLS

- **Languages & Data:** Python (Advanced), SQL (Advanced), Go (Intermediate), NumPy, Pandas, Scikit-learn
- **ML & Forecasting:** PyTorch, TensorFlow, LSTMs, Time Series Modeling, Probabilistic Models, sktime, LLMs, RAG, LangChain, LangGraph
- **MLOps & Cloud:** Docker, Kubernetes, GCP, Vertex AI, Vector Databases, CI/CD, FastMCP
- **Specializations:** Predictive Modeling, Multi-Agent Systems, LLM Fine-tuning, Multimodal Learning, Anomaly Detection

PROJECTS

Agentic Forecaster for sktime | [Github](#) | [Demo](#)

- Drop-in sktime forecaster using a ReAct agent loop to automatically select, fit, and score time series models from plain English descriptions – covering ARIMA, Prophet, ExponentialSmoothing, Theta, TBATS, and more.
- Constrained to 6 auditable tools (no free-form code execution); every decision logged in a full transcript for reproducibility and inspection.
- Supports Claude, GPT-4o, and Gemini backends; Anthropic prompt caching cuts per-step token cost by 60–80%; dual in-process and MCP transport modes enable direct use from Claude Desktop and Cursor.

σ -RAG: Significance-Threshold Retrieval | [Github](#)

- Built and published to PyPI - an open-source RAG library that fits a Gaussian noise floor from cross-document embedding pairs and gates retrieval at a configurable σ threshold; achieves 100% out-of-domain query suppression at 2σ with no loss on answerable queries.
- Reduces average chunks passed to the LLM from 3 (standard top-k) to 1.8 on answerable queries and 0 on unanswerable ones; pure-NumPy core with no dependencies beyond numpy.

WORK EXPERIENCE

AI Engineer | *Beehive AI, Inc.*

Jul 2024 – Present

- Engineered a production-grade agentic RAG system using LangChain and Postgres vector search, enabling natural language to SQL generation for real-time customer analytics.

- Developed an LLM-powered QA framework for data categorization, achieving 80% automation by validating and refining category labels, reducing manual effort and improving classification accuracy.
- Built a scalable virtual persona simulation framework using LLMs, contextual retrieval, and memory systems to model user behavior and drive data-informed product decisions.
- Architected a multi-agent commerce system using Vertex AI, LangGraph, and semantic search, integrating dynamic UI and conversational interfaces for personalized product recommendations.

AI Research Scientist | *Pocket FM*

Jan 2024 – Jul 2024

- Delivered an LSTM-based sequential prediction model leveraging episode-level engagement data and metadata to forecast long-term content performance.
- Led development of a comment ranking system using semantic embeddings, LLM summaries, and engagement signals, resulting in +4.5% CTR and +4.0% increase in comment volume.

AI Fellow | *PI School, Rome, Italy*

Dec 2023 – Feb 2024

- Automated reliability assessment of medical research papers using LLM-based algorithms targeting key quality metrics based on predefined categories.
- Improved GPT-4 evaluation accuracy to 75% through prompt optimization and enhanced Mixtral performance to 69% using Quantized LoRA fine-tuning.

Research Scientist Intern | *Deepgram, Inc.*

Jun 2023 – Sep 2023

- Trained a self-supervised audio embedding model on 35K+ hours of audio data, achieving 86% accuracy across downstream tasks including language detection and domain classification.
- Benchmarked multimodal representations against BERT-based text embeddings, outperforming text-only baselines by 5 percentage points.
- Engineered a multimodal embedding space aligning audio and text representations, enabling cross-modal tasks including speech recognition and translation.

EDUCATION

Ph.D., Physics | *University of California, Riverside*

Sep 2017 – Jun 2024

- Specialized in theoretical and computational physics; developed strong foundations in statistical modeling, probabilistic reasoning, and experimental design.

BS-MS Dual Degree, Physics | *IISER, Kolkata*

Aug 2012 – May 2017